

Practice Exercise: Advanced Generative AI in Data Science Workflow

This practice exercise builds upon the concepts of Generative AI in data science. You will explore practical applications of GenAI for sales forecasting, exploratory data analysis, synthetic data generation, natural language to SQL, and automated code documentation.

Exercise 1: Sales Forecasting Insights with GenAI

Scenario: You are tasked with analyzing retail sales data and providing forecasts.

Tasks:

1. Upload 'SalesData.csv' to a GenAI tool such as ChatGPT (Advanced Data Analysis) or Gemini.
2. Generate top 3 insights on product sales trends, regional demand, and revenue contribution.
3. Identify anomalies such as unusual spikes or drops in sales across months.
4. Request a 3-month sales forecast and plot monthly sales trends.
5. Summarize the results into a plain English business report.

Exercise 2: EDA Summarization with GenAI

Scenario: You are exploring a customer dataset and need to quickly summarize insights.

Tasks:

1. Load the Titanic dataset (or any dataset of your choice).
2. Use a GenAI tool to generate a natural language summary of key insights such as survival rates, demographics, or any notable patterns.
3. Compare AI-generated insights with your own EDA summary.
4. Discuss where GenAI helps and where manual analysis is still important.

Exercise 3: Synthetic Data Generation

Scenario: You want to generate customer data for testing a retail model.

Tasks:

1. Define schema: id, age, gender, income, purchases.
2. Use GenAI to generate at least 10 rows of synthetic customer data.
3. Compare this with synthetic data generated by the Faker library in Python.
4. Evaluate the realism and usefulness of AI-generated data.

Exercise 4: Natural Language to SQL

Scenario: You are analyzing retail transactions but do not know SQL.

Tasks:

1. Present schema: customers(id, name, city), purchases(id, customer_id, amount, date).
2. Use GenAI to generate SQL queries such as:

- Average spend by city.
 - Top 5 customers by spending.
 - Total monthly revenue.
3. (Optional) Run the queries on a sample database and validate results.

Exercise 5: Auto-Documentation of Code

Scenario: You need to create documentation for a machine learning pipeline.

Tasks:

1. Take a sample ML pipeline code (data loading → preprocessing → model training).
2. Ask a GenAI tool to explain the code in simple terms.
3. Compare AI-generated explanations with your own manual documentation.
4. Reflect on how AI can speed up technical documentation.

Reflection

After completing the exercises, reflect on:

- How GenAI tools improve speed and accessibility in data science workflows.
- Scenarios where AI complements vs. replaces traditional approaches.
- Risks of relying solely on AI for analysis and documentation.